

Risk Scoring Framework

1. Business Question

Which customer segments have the highest loan default probability, and which applications should the credit team manually review?

2. Purpose

A rule-based decision-support score that summarizes historical risk indicators for each loan application and supports manual-review prioritization in Power BI.

3. Scoring Principles

- Points are based on observed historical default rates in the cleaned dataset.
- Higher points represent higher observed risk within a factor.
- Factors with weak differentiation receive limited weight.
- Small-sample categories are interpreted cautiously.
- The score is not a machine-learning prediction or guaranteed future default probability.

4. Overall Baseline

Observed default rate in the cleaned dataset: approximately 11.94%.

5. Factor Weighting

Factor	Point Range
Grade	0–5
FICO	0–2
Interest Rate	0–3
Term	1–2
Verification Status	0–2
Income	0–2
Home Ownership	0–2
Employment Length	1–2
Purpose	1–2
DTI	1–2
Loan Amount	1

6. Detailed Scoring Rules

Grade

Category / Range	Observed Default Rate	Points
A	3.29%	0

B	7.95%	0
C	13.22%	1
D	18.92%	2
E	26.74%	4
F	34.96%	5
G	38.17%	5

FICO

Category / Range	Observed Default Rate	Points
612–669	17.10%	2
670–739	11.97%	1
740–799	5.14%	0
800+	3.12%	0

Interest Rate

Category / Range	Observed Default Rate	Points
<10%	4.72%	0
10–15%	11.20%	1
16–20%	18.17%	2
20%+	24.16%	3

Term

Category / Range	Observed Default Rate	Points
36 months	10.20%	1
60 months	16.24%	2

Verification Status

Category / Range	Observed Default Rate	Points
Not Verified	8.08%	0
Source Verified	12.36%	1
Verified	15.93%	2

Income

Category / Range	Observed Default Rate	Points
<\$40K	14.41%	2
\$40–65K	13.18%	2
\$65–100K	11.26%	1
\$100–150K	9.27%	1
\$150–250K	8.03%	0
\$250K+	6.71%	0

Home Ownership

Category / Range	Observed Default Rate	Points
MORTGAGE	10.37%	1
OWN	11.85%	1
RENT	13.93%	2
NONE*	16.33%	2
OTHER*	21.71%	2
ANY*	5.70%	0

Employment Length

Category / Range	Observed Default Rate	Points
10+ years	11.14%	1
4 years	11.69%	1
<1 year	11.77%	1
5 years	11.85%	1
6 years	11.88%	1
2 years	11.91%	1
3 years	11.95%	1
1 year	12.35%	2
7 years	12.60%	2
9 years	12.81%	2
8 years	13.22%	2
Unknown	14.59%	2

Purpose

Category / Range	Observed Default Rate	Points
credit_card	9.70%	1
car	9.08%	1
home_improvement	10.40%	1
major_purchase	10.94%	1
vacation	11.27%	1
house	11.50%	1
other	11.89%	1
debt_consolidation	12.96%	1
medical	12.47%	1
wedding	12.20%	1
moving	14.50%	2
renewable_energy*	15.43%	2
small_business	18.97%	2
educational*	20.75%	2

DTI

Category / Range	Observed Default Rate	Points
<20%	10.34%	1
20–35%	14.33%	2
36–43%	15.06%	2
44–50%*	9.25%	1
50%+*	5.84%	1

Loan Amount

Category / Range	Observed Default Rate	Points
\$0–10K	10.44%	1
\$10–20K	13.04%	1
\$20–30K	13.23%	1
\$30–40K	11.71%	1

* Small-sample category; observed rate has been interpreted cautiously.

7. Maximum Score

Maximum possible score = 25 points. Risk Score = sum of points across all 11 factors.

8. Risk Categories

Score	Category	Interpretation
0–6	Low Risk	Relatively lower historical risk indicators
7–12	Moderate Risk	Some elevated risk indicators
13–18	High Risk	Multiple elevated risk indicators
19–25	Very High Risk	Many elevated risk indicators

9. Manual Review Rule

Initial project rule: Risk Score $\geq 7 \rightarrow$ Manual Review = Yes. Scores below 7 are not automatically flagged.

This is a rule-based project decision, not a validated credit-policy cutoff.

10. Dashboard Use

The Power BI Risk Scoring Dashboard will show Applicant ID, Risk Score, Risk Category, Manual Review flag, factor-level point contributions, and key application attributes.